Octal N.S.

HexaDecimal N.S

Algorithms & flowcharts

0, 1, 2, 3, 4, 5, 6, 7

Base: 8

$(1237)_8$ ✓

$(9870)_8$ X

$(32)_8 = (26)_{10}$

$8^1 \quad 8^0$

$8 \quad 1$

$24 + 2$

26

$(171)_8 = (121)_{10}$

$8^2 \quad 8^1 \quad 8^0$

$64 \quad 8 \quad 1$

$64 + 56 + 1$

121

Octal N.S.

HexaDecimal N.S

Algorithms & flowcharts

0,1,2,3,4,5,6,7

Base: 8

$(1237)_8$ ✓

$(9870)_8$ X

$(73)_8 = (111011)_2$

$8^1 \quad 8^0$

$56 + 3$

$(59)_{10}$

-32

27

-16

11

-8

3

-2

1

-1

0

$(1\ 1\ 1\ 0\ 1\ 1)$

$32\ 16\ 8\ 4\ 2\ 1$

(73)

(7)

111

$(531)_8 =$

$101 \quad 011$

$(262)_8 = ($

$010 \quad 110$

6, 6, 7

✓

✗

$(32)_8 = (26)_{10}$

$8^1 \quad 8^0$

$8 \quad 1$

$24 + 2$

26

$(171)_9 = (121)_{10}$

$9^2 \quad 9^1 \quad 9^0$

$64 \quad 9 \quad 1$

$64 + 56 + 1$

121

$(93)_{10} = (135)_8$

$8^2 \quad 8^1 \quad 8^0$

$64 \quad 8 \quad 1$

$64 + 24 + 5$

93

8	93	
8	11	5
8	1	3
	0	1
		135

$(77)_{10} = (115)_8$

$64 + 8 + 5$

77

8	77	
8	9	5
8	1	1
	0	1

$= (111011)_2$

1 1 0 1 1
16 8 4 2 1

2^3

$(73)_8 = (111011)_2$

(7) (3)

111 011

2	7
2	3 1
2	1 1
	0 1

2	3
2	1 1
	0 1

11

2	5
2	2 1
2	1 0
	0 1

$(531)_8 = (101011001)_2$

101 011 001

$(262)_8 = (010110010)$

010 110 010

2^3

$(73)_8 = (111011)_2$

(7) (3)

111 011

$(531)_8 = (101011001)_2$

101 011 001

$(262)_8 = (10110010)$

010 110 010

2	7	
2	3	1
2	1	1
	0	1

2	3	
2	1	1
	0	1

11

2	5	
2	2	1
2	1	0
	0	1

Octal N.S

HexaDecimal N.S

Algorithms & flowcharts

0,1,2,3,4,5,6,7

Base: 8

$(1237)_8$ ✓

$(9870)_8$ X

0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

$(101110)_2 = (56)_8$

$32 \times 1 + 4 \times 1$

$(46)_{10}$

8	46	
8	5	6
	0	5
		5

101110

5 6 $_8$

$$(\underbrace{111}_{7}\underbrace{011}_{3})_2 = (73)_8$$

$$(\underbrace{101}_{5}\underbrace{011}_{3}\underbrace{001}_{1})_2 = (531)_8$$

$$(\underbrace{001}_{1}\underbrace{001}_{1}\underbrace{010}_{2})_2 = (112)_8$$

0
1
2
3
4
5
6
7
8
9
10 → A
11 → B
12 → C
13 → D
14 → E
15 → F

Base: 16

$(15)_{16}$

$(F)_{16}$

$(12Y)_{16}$

$(12)_{16} = (18)_{10}$

$16^1 \quad 16^0$

$16 + 2$

18

$(A1)_{16} = (161)_{10}$

$16^2 \quad 16^0$

$160 + 1$

Octal N.S.

Hexadecimal N.S

Equations & flowcharts

8 2^3

16 2^{4}

0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111
8	1000
9	1001
10 → A →	1010
11 → B	1011
12 → C	1100
13 → D	1101
14 → E	1110
15 → F	1111

Base: 16

$(15)_{16}$

$(F)_{16}$

$(12Y)_{16}$

$(A2)_{16} = (10100010)_2$

$\frac{16^1}{160}$ $\frac{16^0}{2}$

$\boxed{162}$

1	0	1	0	0	0	1	0
~~256~~ 128	64	32	16	8	4	2	1

162
128
34
32
2
2
0

$(A2)_{16} = (10100010)_2$

$\boxed{1010}$ $\boxed{0010}$

$(E5)_{16} = (11100101)$

1110 0101

$(ABC)_{16}$

$\boxed{1010}$ $\boxed{1011}$ $\boxed{1100}$

2 | 16
2 | 5 0
2 | 2 1
2 | 1 0
0 1

$(8A)_{16} = (10001010)_2$

$(5E)_{16} = (101\ 1110)$

Octal N.S.

Hexadecimal N.S

Algorithm & flowcharts

$8 \quad 2^3$

$16 \quad 2^4$

0	
1	
2	
3	
4	
5	
6	111
7	1000
8	1001
9	1010
10 → A →	1011
11 → B	1100
12 → C	1101
13 → D	1110
14 → E	1111
15 → F	

Base 16

$(15)_{16}$

$(F)_{16}$

$(12Y)_{16}$

$(A2)_{16} = (10100010)_2$

$16^1 \quad 16^0$

$160 \quad 2$

$\boxed{162}$

1	0	1	0	0	0	1	0
~~256~~ 128	64	32	16	8	4	2	1

$(A2)_{16} = (10100010)_2$

$\boxed{1010} \quad \boxed{0010}$

162
128
34
32
2
2
$\boxed{0}$

$(263)_{10} = (107)_{16}$

16	263	7
16	16	0
16	1	1
	0	

$16^2 \quad 16^1 \quad 16^0$

$256 + 0 + 7$

$(263)_{10}$

$(164)_{10} = (A4)_{16}$

16	164	
16	10	4
	0	10

10 4

A 4

$(A2)_{16} = (10100010)_2$

10 2

A 2

$(E5)_{16} = (1110\ 0101)$

1110 0101

E 5

$(ABC)_{16}$

1010 1011 1100

10 11 12

A B C

$(8A)_{16} = (10001010$

$(5E)_{16} = (101\ 111$

$(A2)_{16} = (10100010)_2$

10 2

A 2

$(E5)_{16} = (11100101)$

1110 0101

14 5

E 5

$(10101)_2 = (\quad)_{16}$

$(ABC)_{16}$

1010 1011 1100

10 11 12

A B C

2	160	
2	5	0
2	2	1
2	1	0
	0	1

$(8A)_{16} = (10001010)_2$

$(5E)_{16} = (\cancel{0}101\ 1110)$

$(73)_8 = (111011)_2$

(7)	(3)
111	011

$(531)_8 = (101011001)_2$

101 011 001

$(202)_8 = (010110010)$

010 110 010

0,1,2,3,4,5,6,7

Base 8

$(1237)_8$ ✓

$(9870)_8$ X

$(73)_8 = (111011)_2$

$8^1 \quad 8^0$

$56 + 3$

$(59)_{10}$

-32

27

-16

11

-8

3

-2

1

0

$(1 \ 1 \ 1 \ 0 \ 1 \ 1)$

$32 \ 16 \ 8 \ 4 \ 2 \ 1$

(2)

$(73)_8 = (111011)_2$

(7) (3)

111 011

$(531)_8 = (101011001)_2$

101 011 001

$(262)_8 = (010110010)$

010 110 010

$2 \mid 7$

$2 \mid 3 \ 1$

$2 \mid 1 \ 1$

$\quad 0 \ 1$

$2 \mid 3$

$2 \mid 1 \ 1$

$\quad 0 \ 1$

11

$2 \mid 5$

$2 \mid 2 \ 1$

$2 \mid 1 \ 0$

$\quad 0 \ 1$

Octal NS

Hexadecimal NS

Position of Symbols

0 1 2 3 4 5 6 7

Base 8

$(1237)_8$ ✓

$(9870)_8$ ✗

$(94)_8 = (26)_{10}$

$(195)_8 = (135)_{10}$

$(194)_8 = (121)_{10}$

$(77)_8 = (115)_{10}$

Octal N.S ✓

HexaDecimal N.S ✓

Algorithm & flowchart,

8 2^3

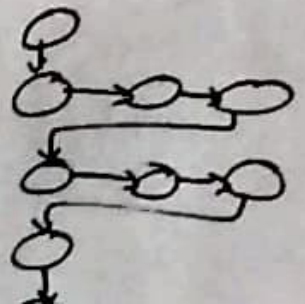
16 2^4

Psudocode

1- user enters height & width

2- Calculate area using height & width

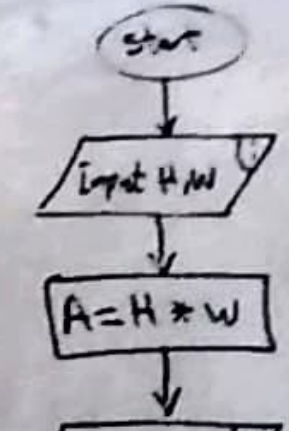
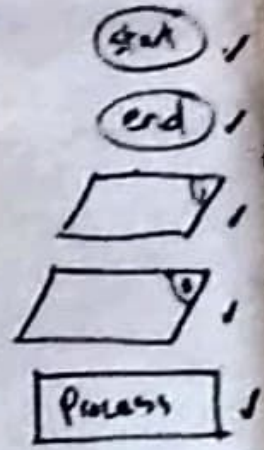
3- print to user the result.



Algorithm

1- Input H, W

2- A = H * W

3- Print A

Flowchart